

# Knowledge Distillation with Hybrid Soft–Label Losses for Imbalanced Lung Nodule Classification

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**Abstract**—Knowledge Distillation (KD) is widely used to compress deep neural networks while maintaining high performance. Loss functions help in choosing the best strategy for image classification. This paper investigates hybrid distillation loss applied in KD for lung nodule malignancy classification on the unbalanced LIDC-IDRI dataset. We propose three hybrid distillation loss functions that combine MSE, Focal Loss, and MSE+Focal Loss for comparison with the baseline loss, using softened teacher outputs to mitigate class imbalance inherent in medical scenarios. Five teacher models (ResNet-50, DenseNet-121, ViT-Small, Swin-Transformer, MaxViT-Tiny) and two lightweight students (EfficientNet-B0, MobileNetV2) were evaluated across three temperatures ( $\tau$ ) in 120 experiments, reducing the students' parameters by 42–97% relative to their respective teachers. MSE-augmented distillation consistently outperformed the traditional distillation loss function. On the test set, MobileNetV2 achieved the highest gain (+4% AUC) with traditional loss and MSE at  $\tau = 10$ , while EfficientNet-B0 improved by +2.0% with traditional loss, MSE, and Focal loss combination at  $\tau = 10$ . These findings indicate that symmetric penalization in probability space improves robustness under class imbalance. The source Code is available at <https://github.com/graciellafavoreto/hybrid-kd-losses>.

**Index Terms**—Knowledge Distillation, Medical Image Classification, Hybrid Distillation Loss

## I. INTRODUCTION

Lung and bronchus cancer represents 20% of cancer deaths in United States<sup>1</sup>. Lung cancer is the most dangerous form of all cancer, it has the highest occurrence worldwide [1], and early detection is crucial for improving prognosis and treatment outcomes [2]. Although manual diagnosis is time-consuming, labor-intensive, and error-prone, professional experience also affects the accuracy of differentiating benign from malignant nodules [3]. Deep learning (DL) models have substantially advanced medical image classification, including lung Computed Tomography (CT) image evaluation [4]–[6]. DL models have shown superior specificity and sensitivity compared to human experts [7]. However, high-capacity architectures such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) are computationally intensive. Therefore, an effective diagnostic tool must not only

achieve high accuracy but also remain accessible and efficient, enabling deployment in resource-constrained clinical environments [8].

Knowledge Distillation (KD) mitigates the computational cost of large models by transferring knowledge from a high-capacity teacher model to a compact student model [9]. Traditional KD approaches use the Kullback–Leibler (KL) divergence as the distillation loss, which quantifies the dissimilarity between a probability distribution and a reference distribution [10].

The literature indicates that traditional logit-based distillation losses suffer from coupling effects between class probabilities, while the limited incorporation of sample-difficulty or class-importance information may further restrict the diagnostic performance of student models. These limitations are particularly critical in imbalanced medical datasets. Although knowledge distillation (KD) improves student performance compared to traditional training, it exhibits performance gaps between benchmarks and real-world scenarios [11]. Most KD research has been conducted on large-scale natural image datasets with relatively balanced class distributions, such as ImageNet, CIFAR-100, TinyImageNet, and FGVR [11]–[13]. In contrast, medical imaging datasets frequently present class imbalance, which may pose additional challenges for standard distillation objectives.

To address these gaps, we proposed hybrid distillation loss functions that extend KL divergence with: (i) symmetric probability-space regularization via Mean Squared Error (MSE), (ii) focal loss, and (iii) a combined MSE + Focal loss. We verify the performance of our methods by conducting experiments on the LIDC-IDRI dataset using five heterogeneous teacher models and two lightweight students, with three temperature values. The main contributions of this study are:

- Three hybrid distillation loss functions integrating KL divergence with soft-distribution MSE and focal loss to address imbalance-sensitive learning.
- A controlled experimental analysis (120 configurations) spanning diverse teachers, students, and temperature regimes.
- An empirical investigation of temperature–loss interactions and their impact on sensitivity–specificity trade-offs.

<sup>1</sup>National Cancer Institute. *SEER Cancer Statistics Factsheets: Common Cancer Sites*. Bethesda, MD. Available at: <https://seer.cancer.gov/statfacts/html/common.html>

- AUC improvements of up to +4% over traditional KL-based distillation
- Practical guidelines for designing robust knowledge distillation pipelines in imbalanced medical image classification tasks.

## II. BACKGROUND AND RELATED WORK

This section reviews prior work on deep learning for lung image classification and on knowledge distillation to improve model performance.

**Fundamentals of Knowledge Distillation:** Knowledge distillation (KD) transfers knowledge from a large teacher model to a compact student model by minimizing the divergence between their output distributions [9]. A central component is the temperature parameter  $\tau$  (scaling factor), which smooths the teacher’s softmax outputs to reveal inter-class relationships suppressed by hard labels [14]. Over the years, KD methods have evolved to narrow the performance gap between student and teacher models by leveraging both final predictions (known as logit-based distillation) and intermediate feature representations (known as feature-based distillation) [12]. Various works have emerged that transfer knowledge from PET/CT data to CT data, capturing both intra-class and inter-class features [15].

Knowledge distillation learning schemes can be categorized into three types: **offline**, **online**, and **self-distillation**. In offline distillation, a teacher model is first trained and then used to guide the student model. Online distillation improves student performance when a high-capacity teacher is unavailable by training models simultaneously. In self-distillation, the same network serves as both the teacher and the student. [16]

**Advances in Logit-Based Distillation:** Recent works have extended classical KD in different directions. Kim et al. [13] showed that MSE outperforms KL divergence as distillation loss on balanced benchmarks. Cui et al. [10] decomposed the KL divergence  $\mathcal{L}_{KL}$  into a weighted MSE and a soft cross-entropy term, proposing an Improved Kullback–Leibler (IKL) to mitigate asymmetric optimization. Kim et al. [12] introduced a dynamic per-sample temperature mechanism, while Ham et al. [11] calibrated distillation confidence based on per-sample complexity. Despite these advances, most approaches remain restricted to balanced datasets or non-medical scenarios.

**Knowledge Distillation for Lung Image Analysis:** Several works have applied KD to lung image classification. Asham et al. [8] achieved 97.92% accuracy with a 0.63M-parameter student for chest X-ray disease classification, though on balanced data. Kabir et al. [17] and Li et al. [15] applied KD to respiratory disease detection and missing-modality lung cancer classification, respectively, but without distillation for more than one teacher.

Table I summarizes the key methodological dimensions of related work and our proposal. It analyzes prior research against five criteria: non-medical approaches tend to fail in the medical domain and often overlook class imbalance, whereas medical approaches either neglect imbalanced data or limit

distillation to a single-teacher architecture. Our proposal aims to address these gaps by introducing hybrid distillation losses to mitigate class imbalance in medical imaging. Additionally, this study incorporates analyses of temperature  $\tau$ , utilizes multiple heterogeneous architectures in distillation, and examines operation-in-space probability. These methods provide a more calibrated and interpretable transfer of inter-class relationships compared to using raw logits.

TABLE I: Comparison of knowledge distillation approaches across key methodological dimensions.

| Work              | Med. | Imb. | $\tau$ Study | Multi-T | Prob. |
|-------------------|------|------|--------------|---------|-------|
| Kim (2021) [13]   | ×    | ×    | ✓            | ✓       | ✓     |
| Cui (2024) [10]   | ×    | ✓    | ✓            | ✓       | ✓     |
| Kim (2024) [12]   | ×    | ×    | ✓            | ✓       | ×     |
| Ham (2024) [11]   | ×    | ×    | ✓            | ✓       | ✓     |
| Kabir (2024) [17] | ✓    | ✓    | ✓            | ×       | ✓     |
| Li (2024) [15]    | ✓    | –    | ✓            | ×       | ✓     |
| Asham (2025) [8]  | ✓    | ×    | ✓            | ✓       | ✓     |
| <b>Ours</b>       | ✓    | ✓    | ✓            | ✓       | ✓     |

Columns denote: medical imaging domain (**Med.**), explicit handling of class imbalance in the distillation loss (**Imb.**), systematic fixed-temperature sensitivity study ( $\tau$  **Study**), distillation across multiple teacher architectures (**Multi-T**), and distillation performed in probability space rather than raw logit space (**Prob.**). – indicates not identified.

## III. MATERIAL AND METHODS

This section comprises the materials and methods employed to develop the proposal. Figure 1 summarizes the experimental pipeline, in which the teacher and student models take lung nodule imaging as input, compute the softmax, and extract the soft and hard labels. The distillation loss function uses soft labels from the teacher, ranging from traditional loss based on KL divergence to the proposed hybrid MSE-based loss to mitigate the effects of class imbalance in medical settings. The student provides hard labels to compute its loss and combines it with the distillation loss to obtain the total loss, improving classification accuracy. Further subsections describe the materials employed to develop this proposal.

### A. Dataset

This study uses the Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) [18], a public benchmark comprising 1010 chest CT scans annotated by up to four radiologists. This dataset is among the largest and most curated benchmarks for lung nodule detection and segmentation [19]. Each nodule receives an integer malignancy score from 1 (Highly Unlikely) to 5 (Highly Suspicious) per annotator; the median score across annotators is used to define a binary label: benign ( $\bar{y} \leq 2.5$ ) and malignant ( $\bar{y} \geq 3.5$ ). Nodules in the range between these values were excluded to avoid an ambiguous labeling. However, because each patient may contain multiple nodules, we consider labels stratified by patient. When only the presence of a single nodule in the malignant class was sufficient to define the patient as malignant, this was necessary to prevent data leakage within their group. The resulting dataset is  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ , where  $x_i \in \mathbb{R}^{3 \times 224 \times 224}$  denotes the three-slice CT crop and

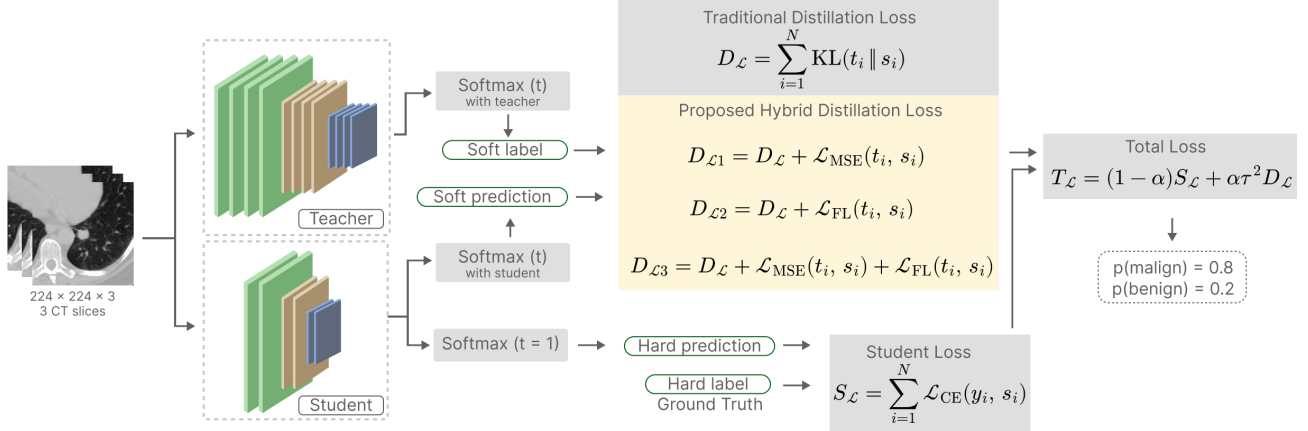


Fig. 1: Experimental pipeline: details teacher–student architectures and the proposed hybrid distillation losses (in light yellow).

$y_i \in \{0, 1\}$  the binary malignancy label, and  $N$  represents the total of nodules.

TABLE II: Characterization of the LIDC-IDRI dataset after pre-processing

| Characteristic                     | Value           |
|------------------------------------|-----------------|
| Total patients                     | 740             |
| Total nodules (after filtering)    | 1695            |
| Benign (malignancy $\leq 2.5$ )    | 1150 (67.8%)    |
| Malignant (malignancy $\geq 3.5$ ) | 545 (32.2%)     |
| Excluded nodules (ambiguous)       | 956             |
| Annotators per nodule (median)     | 3               |
| Median malignancy — Benign         | $1.82 \pm 0.56$ |
| Median malignancy — Malignant      | $4.12 \pm 0.55$ |

## B. Pre-processing

*a) Nodule extraction:* For each annotated nodule, inter-annotator consensus was established at the 50% agreement level (`clevel=0.5`) using the `pylidc` library [20]. The spatial centroid ( $i^*, j^*, k^*$ ) was computed from the consensus bounding box. A  $128 \times 128$  pixel region of interest (ROI) was extracted from the axial plane, centered on ( $i^*, j^*$ ) of slice number  $k^*$  with boundary-aware clipping to prevent out-of-bounds crops. Three consecutive axial slices were selected symmetrically around  $k^*$  (i.e.,  $k^* - 1, k^*, k^* + 1$ ). Figure 2 shows a 2D slice representation, where each one is upscaled to  $224 \times 224$  pixels via bicubic interpolation (order 3), yielding a final representation per nodule of  $V \in \mathbb{R}^{224 \times 224 \times 3}$ .

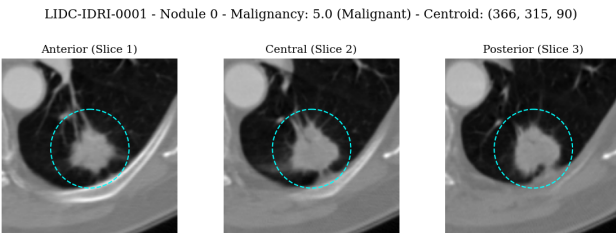


Fig. 2: Pre-processing example

*b) Hounsfield Unit (HU) normalization and ImageNet standardization:* HU values were clipped to the range  $[-1000, 600]$  to encompass the clinically relevant window

from air to soft tissue, and then linearly scaled to  $[0, 1]$ . Subsequently, channel-wise normalization using ImageNet statistics ( $\mu = [0.485, 0.456, 0.406]$ ,  $\sigma = [0.229, 0.224, 0.225]$ ) was applied to align the input distribution with ImageNet-pretrained backbones.

*c) Data augmentation:* During training, random horizontal and vertical flips (each with  $p=0.5$ ) and random 90, 180, and 270 degree rotations ( $p=0.5$ ) were applied. No augmentations were applied during validations or the test set.

## C. Deep Learning Models

The Deep Learning (DL) models employed in this proposal are divided into teacher and student models.

*1) Teacher Networks:* Teacher networks are large, high-capacity models trained to achieve strong predictive performance on the target task. A good teacher model generally leads to better student performance, making teacher selection an important design choice. Based on this, we employed five heterogeneous teacher models: ResNet-50, DenseNet-121, ViT-Small, Swin-Transformer Base, and MaxViT-Tiny, covering convolutional, transformer-based, and hybrid-attention architectures to assess distillation across diverse representational paradigms. All models were initialized from ImageNet-pretrained weights and fine-tuned for 3-channel input (corresponding to three axial CT slices) and 2-class output (benign/malignant). A dropout rate of 0.3 was applied at the classifier head. Training employed the AdamW optimizer ( $\text{lr} = 10^{-4}$ , weight decay = 0.05) with a cosine-annealed schedule over up to 50 epochs, using early stopping with a patience of 8 epochs, monitored on the validation AUC.

*2) Student Networks:* Student networks are compact models trained to approximate the teacher’s behavior while requiring significantly fewer parameters and computational resources, enabling deployment in latency-sensitive or hardware-constrained environments. Two lightweight CNN architectures were adopted as student models: EfficientNet-B0 and MobileNetV2. Both are designed for computational efficiency, employing compound scaling and depthwise separable convolutions, respectively, making them suitable for deployment in resource-constrained environments. Students were trained for

100 epochs using Adam optimization (learning rate  $1 \times 10^{-4}$ , batch size 16) with early stopping (patience = 8). Distillation was performed under three temperature settings ( $\tau \in 4, 10, 30$ ) and four loss variants: traditional Kullback–Leibler (KL) divergence, KL with Mean Squared Error (MSE) (KL+MSE), KL with Focal Loss (KL+Focal), and KL combined with MSE and Focal Loss (KL+MSE+Focal), including  $\alpha = 0.5$  and focal modulation parameter  $\gamma = 2.0$ .

#### D. Hybrid Knowledge Distillation Loss

We developed a hybrid distillation loss inspired by the traditional KL divergence-based approach. Knowledge distillation [9] transfers the predictive behavior of a large teacher network  $T$  to a slighter student  $S$  by training the student to match the teacher’s softened output distribution. Let  $z^T$  and  $z^S$  denote the teacher and student logits, respectively. Also,  $t_i = \text{softmax}(z_i^T/\tau)$  and  $s_i = \text{softmax}(z_i^S/\tau)$  are the temperature-scaled soft distributions of the teacher and student, respectively, and the distillation loss is given by:

$$D_{\mathcal{L}} = \sum_{i=1}^N \text{KL}(t_i \| s_i) \quad (1)$$

$\tau$  is the temperature parameter, which is set to higher values to yield softer probability distributions that reveal more inter-class relational information from the teacher. Distillation loss is calculated using the values of output in the softmax layer between the teacher and student networks. Total Loss ( $T_{\mathcal{L}}$ ), where  $S_{\mathcal{L}}$  represents the loss applied in the student model and  $D_{\mathcal{L}}$  represents the distillation loss between teacher and student, as follows:

$$T_{\mathcal{L}} = (1 - \alpha)S_{\mathcal{L}} + \alpha\tau^2 D_{\mathcal{L}} \quad (2)$$

Although MSE directly penalizes magnitude differences between logits, it is sensitive to noisy labels at low  $\tau$ , extreme logit matching can destabilize training, whereas KL divergence mitigates this problem by operating over normalized distributions [13]. Conversely, KL divergence alone may underpenalize large prediction discrepancies when class distributions are well separated. To exploit the complementary strengths of both losses, we propose the three hybrid loss variations below that augment KL divergence with MSE and Focal Loss, the latter up-weighting hard samples that the averaged KL term tends to vanish:

$$D_{\mathcal{L}1} = \sum_{i=1}^N \text{KL}(t_i \| s_i) + \mathcal{L}_{\text{MSE}}(t_i, s_i) \quad (3)$$

$$D_{\mathcal{L}2} = \sum_{i=1}^N \text{KL}(t_i \| s_i) + \mathcal{L}_{\text{FL}}(t_i, s_i) \quad (4)$$

$$D_{\mathcal{L}3} = \sum_{i=1}^N \text{KL}(t_i \| s_i) + \mathcal{L}_{\text{MSE}}(t_i, s_i) + \mathcal{L}_{\text{FL}}(t_i, s_i) \quad (5)$$

$\mathcal{L}_{\text{MSE}}$  and  $\mathcal{L}_{\text{FL}}$  are computed over these distributions rather than raw logits. Note that MSE represents Mean Squared Error and  $F_{\mathcal{L}}$  represents the focal loss.

## IV. EXPERIMENTS, RESULTS, AND DISCUSSION

### A. Experimental setup

We conducted the experiments on a server equipped with an NVIDIA A100-SXM4 GPU with 40 GB of memory, CUDA-compatible, providing high performance for deep neural network training, a total RAM of approximately 541 GB, and a Linux operating system. We developed the code in Python 3.11 using PyTorch 2.10.0 with CUDA 13.0, Timm 1.0.25 for model architectures, scikit-learn 1.8.0 for evaluation and utilities, and standard scientific libraries (NumPy, Pandas, Matplotlib). We also split the dataset using a stratified random 5:1:1 split for training, testing, and validation.

### B. Evaluation Metrics

The performance evaluation of teacher and student models comprises traditional metrics adopted in CT image classification studies [2], [4], [21], including Area Under the ROC Curve (AUC), Accuracy (ACC), Sensitivity, Specificity, and F1-score. AUC measures the model’s ability to discriminate between classes. ACC quantifies the overall proportion of correctly classified samples. Sensitivity (true positive rate) evaluates the ability to correctly identify positive cases, while Specificity (true negative rate) measures the correct identification of negative cases. The F1-score, defined as the harmonic mean of precision and recall, provides a balanced assessment of performance, particularly in the presence of class imbalance.

### C. Teacher Model Performance

First, we evaluated the teacher’s model performance to achieve the best performance on lung nodule imaging classification, and then built a larger model that transfers knowledge under imbalanced data. Table III presents the validation and test results. Transformer-based models outperformed CNNs overall in the validation set, where ViT-Small achieved the highest AUC (96.36%) and sensitivity (96%) with strong capability for malignancy classification. Swin-Transformer addressed the highest performance overall, with accuracy (90.95%), specificity (92.86%), and F1 Score (85.53%). ResNet50 achieved the highest AUC (88.84%) in the test set and Swin-Transformer achieved the strongest overall classification performance, with the highest accuracy (84.44%), specificity (93.45%), and F1-score (75.00%), whereas ViT-Small had the highest sensitivity (77.53%).

From a clinical perspective, higher sensitivity enhances the detection of malignant nodules, while higher specificity reduces false positives at the expense of potentially missing malignant cases. Considering the trade-off between sensitivity and specificity, ViT-Small yields favorable results due to strong AUC and discriminative performance across datasets. Note that teachers model transfer knowledge to improve students’ performance.

### D. Student Model Performance

The student models provide lightweight architectures that leverage knowledge from teacher networks. Table IV shows

TABLE III: Teacher Models – Validation and Test (%)

| Model       | Validation (%) |              |              |              |              | Test (%)     |              |              |              |              |
|-------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             | AUC            | Acc          | Sens         | Spec         | F1           | AUC          | Acc          | Sens         | Spec         | F1           |
| ResNet50    | 93.72          | 88.07        | 88.00        | 88.10        | 81.99        | <b>88.84</b> | 80.93        | 73.03        | 85.12        | 72.63        |
| DenseNet121 | 94.10          | 87.24        | 86.67        | 87.50        | 80.75        | 86.51        | 80.54        | 73.03        | 84.52        | 72.22        |
| ViT-S       | <b>96.36</b>   | 88.48        | <b>96.00</b> | 85.12        | 83.72        | 87.47        | 80.93        | <b>77.53</b> | 82.74        | 73.80        |
| Swin-T      | 95.86          | <b>90.95</b> | 86.67        | <b>92.86</b> | <b>85.53</b> | 86.96        | <b>84.44</b> | 67.42        | <b>93.45</b> | <b>75.00</b> |
| MaxViT-T    | 92.75          | 87.65        | 82.67        | 89.88        | 80.52        | 86.89        | 82.49        | 69.66        | 89.29        | 73.37        |

EfficientNet-B0 and MobileNetV2 significantly reduce model complexity, containing 4.01M (15.3 MB) and 2.23M (8.5 MB) parameters, respectively, compared to teacher models ranging from 6.96M to 86.7M parameters and up to 330.9 MB. These results highlight the substantial reduction in computational and storage requirements achieved through knowledge distillation.

TABLE IV: Parameter count and model size for teacher and student

| Role    | Model            | Params (M) | Size (MB) |
|---------|------------------|------------|-----------|
| Teacher | ResNet-50        | 23.5       | 89.7      |
|         | DenseNet-121     | 6.96       | 26.5      |
|         | ViT-Small        | 21.7       | 82.7      |
|         | Swin-Transformer | 86.7       | 330.9     |
|         | MaxViT-Tiny      | 30.4       | 116.0     |
| Student | EfficientNet-B0  | 4.01       | 15.3      |
|         | MobileNetV2      | 2.23       | 8.5       |

Figure 3 summarizes the test AUC for each teacher–student pair under different distillation loss functions and temperature values ( $\tau \in \{4, 10, 30\}$ ). Columns correspond to teacher models and rows to student architectures. Red arrows indicate the best AUC among all losses, while the blue column in each subplot represents the baseline KL distillation loss ( $\mathcal{L}_{KL}$ ), used for comparison in medical imbalanced data. Overall, hybrid functions ( $\mathcal{L}_{KL} + \text{MSE}$ ,  $\mathcal{L}_{KL} + \text{Foc}$ , and  $\mathcal{L}_{KL} + \text{MSE} + \text{Foc}$ ) achieved the best performance in most cases. These findings indicate a consistent advantage over the classical KL formulation without any additional term.

Based on students’ architectures, EfficientNet-B0 achieves the best AUC of 88.1% with MaxViT-T, compared to the best baseline with DenseNet-121, improving by about 1.2%. Meanwhile, MobileNetV2 achieves the best AUC of 88.6% with ResNet-50, a gain of 3.5% to the best value of  $\mathcal{L}_{KL}$  achieved by MaxViT-T. In our analyses, transformer-based teachers provide the greatest improvements in general highest values, whose softer probability distributions benefit more from auxiliary loss terms, while CNN-teachers show smaller but still consistent gains. Despite using the same teacher, the highest gains were observed by CNN-based networks.

Table V shows that  $\mathcal{L}_{KL+\text{MSE}}$  yields the best result for the teacher in both students, once the MSE term appears in all occurrences of the best hybrid loss. Traditional KL-loss can be sensitive to high values of  $\tau$ , showing better results in most cases when  $\tau = 4$ . In contrast,  $\mathcal{L}_{KL+\text{MSE}}$  presents the highest AUC across all temperature values, demonstrating more consistent results.

Figure 3 and Table V demonstrate that hybrid distillation losses outperform the  $\mathcal{L}_{KL}$  (baseline method) across all teacher–student- $\tau$  combinations. These findings help to validate that the traditional KL distillation approach lacks the ability to handle imbalanced datasets and medical scenarios with high similarity.

TABLE V: Best test AUC (%) of hybrid losses vs.  $\mathcal{L}_{KL}$  baseline per teacher–student pair.  $\Delta$  denotes the absolute gain.  $\tau^*$  is the temperature yielding the best result.

| Teacher      | Student         | $\mathcal{L}_{KL}$ | $\tau_{KL}^*$ | Best Hybrid | Best Loss                                | $\tau_{\text{hyb}}^*$ | $\Delta$    |
|--------------|-----------------|--------------------|---------------|-------------|------------------------------------------|-----------------------|-------------|
| ResNet-50    | EfficientNet-B0 | 85.6               | 30            | 87.6        | $\mathcal{L}_{KL+\text{MSE}+\text{Foc}}$ | 10                    | +2.0        |
|              | MobileNetV2     | 84.6               | 4             | 88.6        | $\mathcal{L}_{KL+\text{MSE}}$            | 10                    | +4.0        |
| DenseNet-121 | EfficientNet-B0 | 87.3               | 10            | 88.0        | $\mathcal{L}_{KL+\text{MSE}}$            | 4                     | +0.7        |
|              | MobileNetV2     | 83.6               | 10            | 85.9        | $\mathcal{L}_{KL+\text{MSE}}$            | 10                    | +2.3        |
| ViT-Small    | EfficientNet-B0 | 85.5               | 4             | 85.6        | $\mathcal{L}_{KL+\text{MSE}+\text{Foc}}$ | 30                    | +0.1        |
|              | MobileNetV2     | 81.4               | 30            | 84.1        | $\mathcal{L}_{KL+\text{MSE}}$            | 4                     | +2.7        |
| Swin-T       | EfficientNet-B0 | 85.1               | 4             | 86.4        | $\mathcal{L}_{KL+\text{MSE}}$            | 10                    | +1.3        |
|              | MobileNetV2     | 82.2               | 30            | 84.2        | $\mathcal{L}_{KL+\text{MSE}}$            | 4                     | +2.0        |
| MaxViT-T     | EfficientNet-B0 | 86.9               | 4             | 88.1        | $\mathcal{L}_{KL+\text{MSE}}$            | 30                    | +1.2        |
|              | MobileNetV2     | 85.1               | 10            | 87.0        | $\mathcal{L}_{KL+\text{MSE}}$            | 30                    | +1.9        |
| Average      |                 | 84.7               | –             | 86.5        | –                                        | –                     | <b>+1.8</b> |

## V. CONCLUSION

We proposed hybrid distillation loss functions to provide the best performance under imbalanced data. The results indicate that the distillation loss significantly impacts performance under class imbalance. Traditional KL distillation loss consistently underperformed, while the addition of MSE provided the most robust and stable improvements, achieving a better sensitivity-specificity trade-off. In contrast, incorporating Focal Loss with soft targets reduced consistency and often degraded results, whereas the full combination (KL+MSE+Focal) achieved the highest peak AUC 87.6% at the cost of greater variance. Regarding temperature,  $\tau = 10$  yielded the most reliable performance, whereas excessive smoothing ( $\tau = 30$ ) harmed discriminative capacity. Overall, KL+MSE with moderate temperature emerged as the most reliable distillation strategy in this setting.

This study proposed student models that reduce model parameters by 42-97% relative to their teachers while maintaining competitive AUC, enabling deployment in resource-constrained clinical environments. The results also suggest that CNN-based teachers transfer richer knowledge under a hybrid distillation loss, motivating further investigation into CNN- and ViT-based distillation pipelines for medical imaging.

Finally, limitations include evaluation on a single dataset and the use of fixed  $\alpha$  and  $\gamma$  hyperparameters. Future work will explore joint hyperparameter optimization, dynamic temperature scheduling, and extension to multi-class classification.

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## REFERENCES

- [1] R. V. Adiraju and S. Elias, “A survey on lung ct datasets and research trends,” *Research on Biomedical Engineering*, vol. 37, no. 2, pp. 403–418, 2021.

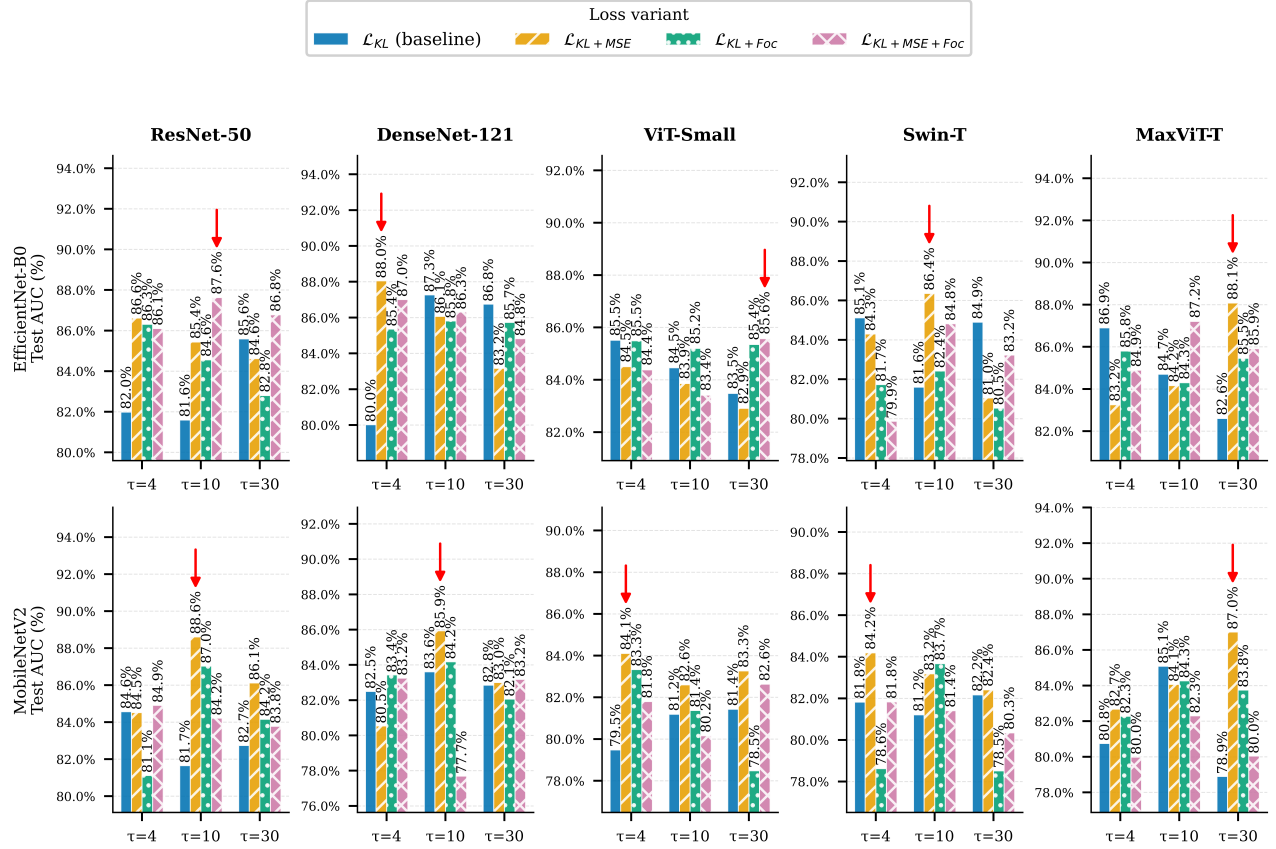


Fig. 3: Test AUC for each teacher–student pair under different distillation losses and temperature values ( $\tau \in \{4, 10, 30\}$ ). Columns correspond to teacher models and rows to student architectures. Red arrows highlight the best overall result among all evaluated distillation loss variants for the given model.

- [2] R. Javed *et al.*, “Deep learning for lungs cancer detection: a review,” *Artificial Intelligence Review*, vol. 57, no. 8, p. 197, 2024.
- [3] H.-H. Chang *et al.*, “Pulmonary nodule classification using a multiview residual selective kernel network,” *Journal of Imaging Informatics in Medicine*, vol. 37, no. 1, pp. 347–362, 2024.
- [4] H. Huang *et al.*, “Enhancing lung nodule classification with triple union efficient attention and neural architecture search,” *Journal of Imaging Informatics in Medicine*, pp. 1–15, 2026.
- [5] J. Wei *et al.*, “Deep learning model for diagnosing and classifying subtypes of chronic pulmonary aspergillosis in chest ct,” *Mycoses*, vol. 68, no. 4, p. e70061, 2025.
- [6] F. Wang *et al.*, “Pneumonia-plus: a deep learning model for the classification of bacterial, fungal, and viral pneumonia based on ct tomography,” *European Radiology*, vol. 33, pp. 8869–8878, 12 2023.
- [7] T.-W. Wang *et al.*, “Standalone deep learning versus experts for diagnosis lung cancer on chest computed tomography: a systematic review,” *European Radiology*, vol. 34, no. 11, pp. 7397–7407, 2024.
- [8] M. A. o. Asham, “A lightweight deep learning model with knowledge distillation for pulmonary diseases detection in chest x-rays,” *Multimedia Tools and Applications*, vol. 84, no. 15, pp. 14 885–14 913, 2025.
- [9] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” 3 2015. [Online]. Available: <http://arxiv.org/abs/1503.02531>
- [10] J. Cui *et al.*, “Decoupled kullback-leibler divergence loss,” in *Advances in Neural Information Processing Systems*, A. Globerson *et al.*, Eds., vol. 37. Curran Associates, Inc., 2024, pp. 74 461–74 486.
- [11] G. Ham *et al.*, “Difficulty level-based knowledge distillation,” *Neuro-computing*, vol. 606, p. 128375, 2024.
- [12] S. Kim *et al.*, “Maximizing discrimination capability of knowledge distillation with energy function,” *Knowledge-Based Systems*, vol. 296, p. 111911, 2024.
- [13] T. Kim *et al.*, “Comparing kullback-leibler divergence and mean squared error loss in knowledge distillation,” 08 2021, pp. 2628–2635.
- [14] F. Wang and H. Liu, “Understanding the behaviour of contrastive loss,” *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2495–2504, 2020.
- [15] X. Li, “A neural network based on knowledge distillation for classification of lung cancer with missing modalities,” *SPIE-Intl Soc Optical Eng*, 8 2024, p. 55.
- [16] J. Gou *et al.*, “Knowledge distillation: A survey,” *International journal of computer vision*, vol. 129, no. 6, pp. 1789–1819, 2021.
- [17] M. M. Kabir, Mridha *et al.*, “Detection of COVID-19, pneumonia, and tuberculosis from radiographs using AI-driven knowledge distillation,” *Heliyon*, vol. 10, no. 5, p. e26801, 2024. [Online]. Available: <https://doi.org/10.1016/j.heliyon.2024.e26801>
- [18] S. G. Armato III, G. McLennan *et al.*, “Data from lidc-idri,” 2015. [Online]. Available: <https://doi.org/10.7937/K9/TCIA.2015.LO9QL9SX>
- [19] M. Nan *et al.*, “Mtl-lungnet: A unified model for nodule segmentation and malignancy classification,” in *International Conference on Innovative Techniques and Applications of Artificial Intelligence*. Springer, 2025, pp. 213–226.
- [20] M. C. Hancock and J. F. Magnan, “Lung nodule malignancy classification using only radiologist-quantified image features as inputs to statistical learning algorithms: probing the lung image database consortium dataset with two statistical learning methods,” *Journal of Medical Imaging*, vol. 3, no. 4, pp. 044 504–044 504, 2016.
- [21] Y. Hu *et al.*, “Lung ct-based multi-lesion radiomic model to differentiate between nontuberculous mycobacteria and mycobacterium tuberculosis,” *Medical Physics*, vol. 52, no. 2, pp. 1086–1095, 2025.